

Unity-MCP define matrix checklist

Use this after enabling/disabling optional `FEATURE_*` symbols. Goal: Editor compiles and MCP assemblies load only when their constraints are satisfied.

Always (Unity-MCP installed)

Check	Expect
<code>com.ivanmurzak.unity.mcp</code> present	Package Manager
<code>UNITY_MCP_READY</code> / <code>UNITY_MCP_DEPS_3</code>	Set by DependencyResolver after NuGet restore
<code>jp.kshoji.midi.mcp</code> loaded	<code>midi-features-status</code> → Assemblies

Feature matrix

Define	Runtime asm	MCP asm	Smoke (Edit unless noted)
<i>(none beyond MCP)</i>	—	<code>jp.kshoji.midi.mcp</code>	<code>midi-features-status</code> ; Play: <code>midi-init</code> → <code>midi-devices-list</code>
<code>FEATURE_USE_TIMELINE</code>	<code>jp.kshoji.midi.timeline</code>	<code>jp.kshoji.midi.mcp.timeline</code>	<code>midi-timeline-setup-playback</code>
<code>FEATURE_INPUT_SYSTEM</code>	<code>jp.kshoji.midi.inputsystem</code>	<code>jp.kshoji.midi.mcp.inputsystem</code>	<code>midi-inputsystem-setup-bridge</code>
<code>FEATURE_USE_VISUALSCRIPTING</code>	<code>jp.kshoji.midi.visualscripting</code>	<code>jp.kshoji.midi.mcp.visualscripting</code>	<code>midi-vs-status</code>
<code>FEATURE_SCRIPTABLE_AUDIO</code> (+ Unity 6000.3+)	<code>jp.kshoji.midi.scriptableaudio</code>	Editor <code>jp.kshoji.midi.mcp.scriptableaudio</code> + Runtime <code>*.scriptableaudio.runtime</code>	Editor: <code>midi-sa-bootstrap-metronome</code> ; Player: <code>midi-sa-validate</code> → (Play) <code>midi-sa-transport</code>
<code>FEATURE_CHUNITY</code>	<code>jp.kshoji.midi.chunity</code>	Editor <code>jp.kshoji.midi.mcp.chunity</code> + Runtime <code>*.chunity.runtime</code>	Editor: <code>midi-chunity-setup-bridge</code> ; Player: <code>midi-chunity-validate-scene</code>
<code>FEATURE_USE_MPTK</code>	<code>jp.kshoji.midi.mptk</code>	Editor <code>jp.kshoji.midi.mcp.mptk</code> + Runtime <code>*.mptk.runtime</code>	Editor: <code>midi-mptk-bootstrap</code> ; Player: <code>send-test-note</code> ; Pro+ <code>MPTK_PRO</code> : <code>effects-configure</code> / <code>voice-lifecycle</code> / <code>chord-progression</code>
<code>FEATURE_MIDI_NETWORK</code>	<code>jp.kshoji.midi.net</code>	Editor <code>jp.kshoji.midi.mcp.network</code> + Runtime <code>*.network.runtime</code>	Editor: <code>midi-net-hub-client-setup</code> ; Player: <code>midi-net-discovery-status</code> / <code>midi-net-rtt</code>
<code>FEATURE_MIRROR</code> (+ <code>MIRROR</code>)	<code>jp.kshoji.midi.net.mirror</code>	<i>(bridge via <code>midi-net-bridge-setup</code>)</i>	<code>midi-net-bridge-setup</code> <code>framework=mirror</code>
<code>FEATURE_NETCODE</code> (+ <code>NGO</code>)	<code>jp.kshoji.midi.net.netcode</code>	same	<code>framework=netcode</code>
<code>FEATURE_WSNET2</code> (+ <code>MIDI_HAS_WSNET2</code>)	<code>jp.kshoji.midi.net.wsnet2</code>	same	<code>midi-wsnet2-sync</code> -defines then <code>framework=wsnet2</code>

Define OFF

For each row above: remove the define → corresponding MCP asm must be **not-loaded** in `midi-features-status`, and core `jp.kshoji.midi.mcp` must still compile.

Core tools (no extra FEATURE)

Tool	Mode	Notes
<code>midi2-devices-list</code>	Play	Calls <code>InitializeMidi2</code> when <code>initialize=true</code> (not <code>ReadOnly</code>)
<code>midi2-lifecycle</code>	Play	<code>initialize</code> / <code>terminate</code> / <code>status</code>
<code>midi2-send-ump</code> / <code>midi2-send-channel-voice</code> / <code>midi2-send-system</code> / <code>midi2-send-data</code> / <code>midi2-send-flex-data</code> / <code>midi2-send-stream</code>	Play	Structured + raw
<code>midi2-virtual-device</code> / <code>midi2-monitor-read</code>	Play	
<code>midi-send-message</code> / <code>midi-panic</code> / <code>midi-virtual-device-inject</code> / <code>midi-device-info</code>	Play (info Play)	MIDI 1.0 coverage
<code>ump-sequencer-control</code> / <code>ump-recorder-control</code> / <code>ump-sequence-convert</code>	Mixed	Core UMP Sequence
<code>midi-rtp-session</code> / <code>midi-ble-control</code> / <code>midi-nearby-control</code> / <code>midi2-udp-session</code> / <code>midi-transport-status</code>	Play	Transport
<code>midi-device-selection</code> / <code>midi-latency-calibration</code> / <code>midi-output-routing</code> / <code>midi-clock-*</code> / <code>midi-chord-state</code> / <code>midi-theory-utility</code>	Mixed	Foundation / Gameplay
<code>mpe-zone-status</code>	Edit/Play	After <code>SetupMpeZone</code> / <code>setupIfMissing</code> (not <code>ReadOnly</code> when setup)
<code>midi-ci-discover</code>	Play	Needs IN+OUT; waits for Discovery timeout

Smoke checklist (Cursor / Editor host)

Step	Expect
<code>midi-features-status</code>	Assemblies include <code>jp.kshoji.midi.mcp</code> (+ <code>jp.kshoji.midi.mcp.runtime</code>)
Play: <code>midi-init</code> → <code>midi-devices-list</code> → <code>midi-send-note</code> → <code>midi-monitor-read</code>	Success chain
Play: <code>midi-send-message messageType=TimingClock</code>	Success
Play: <code>midi-panic</code>	Success
Play: <code>midi2-lifecycle action=initialize</code> → <code>midi2-send-channel-voice messageType=noteon</code>	Success when MIDI2 devices exist
Edit: <code>ump-sequence-convert</code> (with sample assets)	Success
Play: <code>midi-transport-status</code>	Success (platform notes OK)
<code>FEATURE_MIDI_NETWORK</code> : <code>midi-net-hub-client-setup</code>	Network asm loads (no CS0104)

Player smoke (Standalone or Android)

Build a **Player** with `MidiMcpRuntimeBootstrap` in a loaded scene (or set `UNITY_MCP_HOST`). Point the AI client `mcp.json` at the **MCP Server on the LAN PC**, not the device IP. Android: ensure **INTERNET** permission and that the device can reach the Server URL/port.

Step	Expect
<code>midi-ping</code>	<code>player=true</code> , <code>aiToolAttrs>=1</code> , <code>hostConfigured=true</code> after Host is set; <code>optionalRuntime</code> lists loaded <code>*.runtime</code> asms
<code>midi-ping</code> with empty Host + Bootstrap in scene	Console shows Host missing <code>[Error]</code> / <code>[Warning]</code> ; ping shows <code>hostConfigured=false</code> + hint
<code>midi-ping</code> / <code>tools/list</code> after IL2CPP	Tools still listed; if <code>aiToolAttrs=0</code> , check <code>Mcp/**/Runtime/link.xml</code> (<code>preserve="all"</code> on each <code>jp.kshoji.midi.mcp*.runtime</code>)
Core: <code>midi-init</code> → <code>midi-devices-list</code> → <code>midi-send-note</code> → <code>midi-monitor-read</code>	Success (device-dependent for hardware MIDI)
Wiring missing	Clear <code>[Error]</code> / <code>EditorWiringRequiredError</code> — wire in Editor first (no Player create/assign)
Optional <code>FEATURE_USE_MPTK</code>	Editor bootstrap first → Player <code>midi-mptk-validate-setup</code> → <code>send-test-note</code> → <code>panic</code>
Optional <code>FEATURE_CHUNITY</code>	Editor <code>setup-bridge</code> → Player <code>validate-scene</code> / <code>diagnostics</code>
Optional <code>FEATURE_SCRIPTABLE_AUDIO</code>	Editor <code>midi-sa-bootstrap-*</code> → Player <code>midi-sa-validate</code> → (Play) <code>midi-sa-transport</code>
Optional <code>FEATURE_MIDI_NETWORK</code>	Editor <code>hub-client-setup</code> → Player <code>discovery-status</code> / <code>rtt</code>

Distribution

MCP stays **same-repository optional** under `Assets/MIDI/Scripts/Integrations/Mcp/` . Do not ship Unity-MCP or NuGet plugins in Asset Store core. Separate UPM packaging remains optional and is not required.

VST host tools (`vst3-*`) remain in [Unity-VST3-Bridge](#) (desktop Player MCP; not an Android MCP target).

Coverage detail: [unity-mcp-api-coverage.md](#).

Sample prompts: [unity-mcp-sample-prompts.md](#).